System Test Plan

for the

KNEAD Example System

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DOCUMENT CHANGE HISTORY

The following table is a simple list of released revisions sent for review. Records of reviews and the review artifacts are saved with reviewer information in the The Balance Projectartifact repository.

Change Record

Date	**Version**	**Author(s)**	**Change Reference**
13 Sep 2025	P1	Vinay Agarwal	Preliminary DRAFT version

Draft P1 Preliminary version of this document.

1. Baseline Document draft 1

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CHAPTER 1

Scope

This document provides the System Test Plan (**STP**) for the Balance System, which is known as Balance System. These engineering tests provide the multistage plan for testing of the Balance System, which follows Appendix A of DOD-STD-2106 (Navy) [**ref˙˙DOD˙STD˙2106˙NAVY**].

1.1 Identification

The Balance System, described in this document shall be known as Balance System version 1.0.

1.2 System Overview

Figure 1 shows the high-level architecture for the Balance System system. This diagram shows the major external interfaces that provide the capabilities of Balance System.

The Balance System is a single player game. The game is a standalone game played on the STM32 board. This system would be a game where the user would have to balance a ball on a LCD screen that is builtin on the STM32 board. The objective of the game is to balance the ball on the screen based on the way the board was tilted. Using an onboard gyroscope that communicates across the board to the MCU though SPI communication. Balance System would keep track of the current position of the ball and where the next updated move is. This helps keep track of the system of where the ball is until a movement has occurred. Balance System shall process at a maximum 180 Hz. This would give the user enough time to process the current angle of the ball and be able to present on the LCD-TFT screen.

1.3 Document Overview

This section provides information about this document's security/privacy considerations, contents, structure, and version information.

1.3.1 Security and Privacy Considerations

This document has been identified as ”Controlled Unclassified Information” (CUI). Please follow the control block on the title page for ownership, creation, category, dissemination, and Point of Contact (**POC**) information. This information should be delineated, per `https://www.dodcui.mil` as:

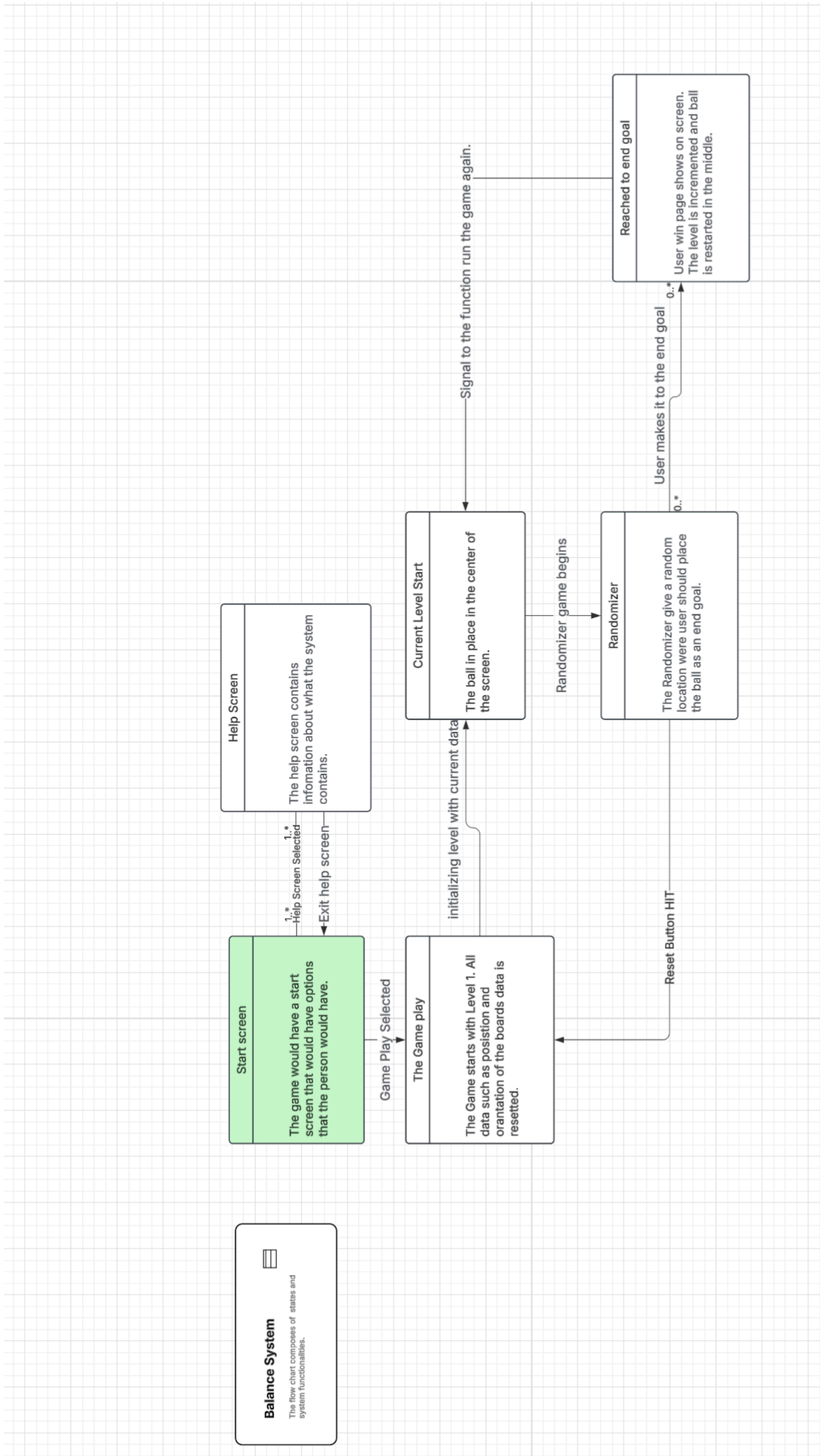

Figure 1: System Overview

Owner the name of the DoD Component (not required if identified in the letterhead)

Creator identification of the office creating the document

Category identification of the categories contained in the document

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1.3.2 Artifact Format

This document format is based upon the guidance in the **STP DID** [**ref˙˙STP˙DID**]. The test planning is documented following the guidelines of ISO-12207 (Software Life Cycle Process) [**ref˙˙ISO˙12207**] and MIL-STD-498 (Software Development and Documentation) [**ref˙˙MIL˙STD˙498**], from which ISO-12207 originated. This document follows the listed **STP** sub-section order.

Section 1 provides an overview of the system and this document.

Section 2 lists general and application-specific reference documents as well as glossary terms and acronyms.

Section 3 summarizes the test environment(s).

Section 4 identifies the tests to be performed.

Section 5 outlines the test schedules.

Section 6 provides any applicable requirement traceability.

Section 7 if needed, lists any general notes as may be applicable.

Appendices if needed, provide additional information as may be needed.

1.3.3 Document Version Information

This document was produced in LaTeX and *BibLaTeX/Biber*. The editing and document preparation were performed using MiKTeX version 2.9 with the build option [LaTeX ⇒ PS ⇒ PDF]. The LaTeX*svn-multi* package was used to glean SVN tracking information, when files are stored in an "SVN" version control system. The style `KNEADdocument` was used to provide the LaTeX and *BibLaTeX/Biber* formatting details.

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CHAPTER 2

References

This section provides a list of referenced items for this document.

2.1 Acronyms and Abbreviations

This section defines acronyms and abbreviations used in this and related documents.

Table 1: Acronym Definitions

Acronym	Definition
YSM	Yourdon Structured Method
LTDC	Layered Transmission Display Controller
STM	STMicroelectronics
MEMS	Micro-Electro-Mechanical Systems
End of acronym definition table	

2.2 Glossary and Definitions

This section defines glossary terms used in this and related documents.

Table 2: Glossary Terms and Definitions

Glossary Term	Definition
STM32F429I	Micro-controller board has all component fit onto one board.
Customer	The professor that is view the grading all assignments.
End of glossary terms table	

2.3 Referenced Documents

This section lists the referenced documents for this document. The references are categorized into two categories:

External Documents not directly associated with this project.

Project Documents that are directly associated with this project.

2.3.1 External Documents

2.3.2 Project Specific Documents

CHAPTER 3

Test Environments

This section describes the test environments to be used for testing the system.

3.1 Test Environment One

The test be taken with the cable plugged in and turned on.

3.1.1 Software Items

The test environment is just needed to have a flat table with a micro usb that is connected to the power source.

3.1.2 Hardware and Firmware Items

The system is needed to have a specific connection that cable that is needed for regular connection.

3.1.3 Other Materials

There is no other materials that is needed for this test.

3.1.4 Proprietary Nature

The proprietary nature, acquirer's rights, and licensing issues for the TEone Test Environment are from the STM32 group.

3.1.5 Installation

The installation issues for the TEone Test Environment are negatable.

3.1.6 Participating Organizations

The participating organizations for the TEone Test Environment are negatable.

3.1.7 Personnel

The personnel planning for the TEone Test Environment is negatable.

3.1.8 Orientation Planning

The orientation planning for the TEone Test Environment is negatable.

3.1.9 Tests to be Performed

The tests to be performed for the TEone Test Environment are to check that the systems are up and running.

3.2 Test Environment Two

The test be taken with the cable plugged in and turned on.

3.2.1 Software Items

The test environment is just needed to have a flat table with a micro usb that is connected to the power source.

3.2.2 Hardware and Firmware Items

The system is needed to have a specific connection that cable that is needed for regular connection.

3.2.3 Other Materials

There is no other materials that is needed for this test.

3.2.4 Proprietary Nature

The proprietary nature, acquirer's rights, and licensing issues for the TEtwo Test Environment are from the STM32 group.

3.2.5 Installation

The installation issues for the TEtwo Test Environment are negatable.

3.2.6 Participating Organizations

The participating organizations for the TEtwo Test Environment are negatable.

3.2.7 Personnel

The personnel planning for the TEtwo Test Environment is negatable.

3.2.8 Orientation Planning

The orientation planning for the TEtwo Test Environment is negatable.

3.2.9 Tests to be Performed

The tests to be performed for the TEtwo Test Environment are to check that the systems are up and running.

CHAPTER 4

Test Identifications

This section defines the tests and test cases to be conducted for validating the Balance System.

4.1 General Information

This section provides general and/or common information for all tests.

4.1.1 Test Levels

The test strategy follows a seven-stage test process, which is defined in the Department of Defense Design Criteria Standard – Development of Shipboard Industrial Test Procedures [**ref˙˙DOD˙STD˙2106˙NAVY**]. The first three stages occur at the component level and the second three stages occur at the assembly, sub-system, and system level. The final stage is complete system testing in its deployed configuration. The test stages are defined as:

1. **Material Receipt Inspection ::** to ensure material received from vendor have received no damage during shipping and parameters match expectations, e.g., part numbers match purchase order information and paperwork matches content. A Return Materiel Authorization (**RMA**) would be started, if required.

2. **Bench Functional Test ::** of components expected configuration (e.g., correct memory, licenses, etc.) and operation as expected in a stand-alone manner (e.g., power-up and basic operational testing).

3. **Installation / Installed Equipment Test ::** of special requirements at the installed component level (e.g., shock, vibration, **EMI**, etc. of **LRUs**).

4. **Intra-system Test ::** of sub-systems basic operation as expected in a stand-alone manner (e.g., **SWaP**, limited functionality, etc.).

5. **Total System Test ::** of the entire system in a test environment that may impart some restrictions on final functionality (e.g. limited communications connections or dummy loads instead of antennas).

6. **System Special Test ::** of special requirements at the sub-system level (e.g., shock, vibration, **EMI**, **EMC**, etc. of entire rack(s) of **LRUs**).

7. **Total Integrated System Tests ::** of the entire system in a deployed environment that imparts no restrictions on final functionality.

4.1.2 Test Classes

Following the seven test stage paradigm, the testing is divided into four classes:

1. **Development Testing** validation of the system components, sub-systems, and system in the development environment, following stages 1-5 above.
2. **Environmental** validation of the system environmental constraints, following stage 6 above.
3. **Design Testing** (**DT**) validation of the system meeting the design specifications, following stage 7 above.
4. **Operational Testing** (**OT**) validation of the system meeting the operational goals and mission needs, following stage 7 above.

4.1.3 Test Conditions

Test conditions will be defined in each test specification (procedure).

4.1.4 Test Progression

The tests will progress in the general order specified above for the stages. This is not to say that all stage-1 tests will be completed before any stage-2 testing begins, and so forth. Additionally, sub-system testing may proceed based on procurement and assembly constraints rather than an order that follows the design hierarchy. It is expected, however, that all developmental, environmental, and design testing will be completed before the final operational testing commences.

4.1.5 Data recording, reduction, and analysis

Each test will define the data to be recorded. The test specifications will define the information needed to perform any analysis needed to validate the specifications covered by each test or test case.

4.2 Planned Tests

A list of test levels and references to a brief description of each test area are:

- **Stage-1** Material Receipt Inspections (§ 4.2.1).

- **Stage-2** Bench Functional Tests (§ 4.2.2).
- **Stage-3** Installation Tests (§ 4.2.3).
- **Stage-4** Intra-system Tests (§ 4.2.4).
- **Stage-5** Total System Tests (§ 4.2.5).
- **Stage-6** System Special Tests (§ 4.2.6).
- **Stage-7** Total Integrated System Tests (§ 4.2.7).

Specific tests and test cases for each area are described in the final subsection for each test level.

4.2.1 Material Receipt Inspections

This section defines the plans for the StageOne validation efforts.

4.2.1.1 Test Objective

The test objective for StageOne is to demonstrate that the received material meets the specifications defined for the material. For equipment items, this means checking part numbers against purchase requisitions and purchase orders to ensure that the correct item, and all expected components of the items, are received. Once items, part numbers, and quantities are verified, the items are checked for any observable damage that may have occurred during shipping. For custom parts, such as machined items or cables, inspections are made to ensure that the parts match appropriate mechanical or electrical characteristics are as defined in the reference design documentation.

Test procedures for these tests may be generic or customized, depending upon the items. Standard parts or equipment, such as **COTS** items, may follow a standardized inspection procedure, with discrepancies noted as needed for the specific parts. Custom items may also use a standardized inspection form, with discrepancies noted against a specific drawing number or other specification artifact. If needed, however, the StageOne may utilize custom inspection procedures to facilitate the necessary level of incoming inspection to ensure quality in the final Balance System.

4.2.1.2 Test Level

The test level for StageOne is **Stage-1** as defined by **ref˙˙DOD˙STD˙2106˙NAVY**, **ref˙˙DOD˙STD˙2106˙NAVY** [**ref˙˙DOD˙STD˙2106˙NAVY**].

4.2.1.3 Test Type or Class

The test type or class for StageOne is development testing.

4.2.1.4 Qualification Method

The test qualification method for StageOne generally is performed via inspection of the received items, but detailed measurements (tests) may be made as necessary for items such as custom machined parts or cable assemblies to ensure they meet the design specifications.

4.2.1.5 Traceability

The test traceability for StageOne often is not included in the overall **RTVM**. Some artifacts of the StageOne inspections may, however, be utilized in higher-level tests. For example, the data sheets for radios may be inspected to complete validation of system-level requirements to meet frequency ranges.

4.2.1.6 Special Requirements

The test special requirements for StageOne are defined as necessary in custom inspection or test procedures.

4.2.1.7 Data Recoding

The test data recoding for StageOne is to record all inspection results in hard copy or electronic form. Electronic form (e.g., a database application with bar code scanning capability) is preferred, but not required, to minimize labor effort and data entry errors.

4.2.1.8 Assumptions or Constraints

The test assumptions or constraints for StageOne are defined in the associated inspection or test procedures.

4.2.1.9 Safety, Security, and Privacy

The test safety, security, and privacy for StageOne are defined in the associated inspection or test procedures. Special care must be taken when test results indicate issues with security measured for the system.

4.2.1.10 Tests and Test Cases

Tests and test cases for this test stage are:

1. **Electronics**
   (a) **Electronics type 1 (e.g., radios)**
   (b) **Electronics type 2 (e.g., computers)**
2. **Mechanical**
   (a) **Mechanical type 1 (e.g., COTS items, e.g., racks, cases, etc.)**

(b) **Mechanical type 2 (e.g., custom items, e.g., metal work)**

4.2.2 Bench Functional Tests

This section defines the plans for the StageTwo validation efforts.

4.2.2.1 Test Objective

The test objective for StageTwo is to demonstrate that the received material operates as expected and includes all necessary components that can best be verified through tests where the equipment is in a powered-on state. These tests ensure that all ordered parts are present and functioning. For example, memory capacity is best verified through a powered-on test. A packing list may state the proper memory is installed, but the memory may not be present or may not be functioning. These tests also ensure that all such items and configurations are accounted for.

Test procedures for these tests may be generic or customized, depending upon the items. Standard parts or equipment, such as **COTS** items, may follow a standardized inspection procedure for each type of equipment, with discrepancies noted as needed for the specific parts. For example, all displays may utilize a common inspection form, with allowances for recording attributes such as display resolution, etc. Custom items, such as a fully assembled printed wiring board (**PWB**) may also use a standardized inspection form, with discrepancies noted against a specific drawing number or other specification artifact. If needed, however, the StageTwo may utilize custom inspection procedures to facilitate the necessary level of bench testing to ensure quality in the final Balance System.

4.2.2.2 Test Level

The test level for StageTwo is **Stage-2** as defined by **ref˙˙DOD˙STD˙2106˙NAVY**, **ref˙˙DOD˙STD˙2106˙NAVY** [**ref˙˙DOD˙STD˙2106˙NAVY**].

4.2.2.3 Test Type or Class

The test type or class for StageTwo is development testing.

4.2.2.4 Qualification Method

The test qualification method for StageTwo generally is performed via demonstration, but detailed measurements (tests) may be made as necessary for items such as **PWB** assemblies to ensure they meet the design specifications.

4.2.2.5 Traceability

The test traceability for StageTwo often is not included in the overall **RTVM**. Some artifacts of the StageTwo inspections may, however, be utilized in higher-level tests. For example, the test results for **PWBs** may be referenced to complete validation of system-level requirements to meet operational constraints.

4.2.2.6 Special Requirements

The test special requirements for StageTwo are defined as necessary in test procedures. Test procedures may include references to overall assembly or operational set-up documentation to prevent duplication of information.

4.2.2.7 Data Recoding

The test data recoding for StageTwo is to record all inspection results in hard copy or electronic form. Electronic form (e.g., a database application with bar code scanning capability) is preferred, but not required, to minimize labor effort and data entry errors.

4.2.2.8 Assumptions or Constraints

The test assumptions or constraints for StageTwo are defined in the associated inspection or test procedures.

4.2.2.9 Safety, Security, and Privacy

The test safety, security, and privacy for StageTwo are defined in the associated inspection or test procedures. Special care must be taken when test results indicate issues with security measured for the system.

4.2.2.10 Tests and Test Cases

Tests and test cases for this test stage are:

1. **Electronics**
   (a) **Electronics type 1 (e.g., radios)**
   (b) **Electronics type 2 (e.g., computers)**
2. **Mechanical**
   (a) **Mechanical type 1 (e.g., COTS items, e.g., racks, cases, etc.)**
   (b) **Mechanical type 2 (e.g., custom items, e.g., metal work)**

4.2.3 Installation Tests

This section defines the plans for the StageThree validation efforts.

4.2.3.1 Test Objective

The test objective for StageThree is to demonstrate one of three distinct areas:

1. **Installation Fit** demonstrates that the equipment can be installed into its designated space. This test is often called the "Form and Fit" validation. Such a validation ensures that the equipment meets the mechanical and electrical interfaces.
2. **Installed Operation** demonstrates that the equipment can be operated once installed into its designated space. This test is often called the "Function" validation. Once the equipment has been shown to capable of installation into its area, this test ensures that the equipment functions as expected. The functionality may include basic operational checks as well as more esoteric assessments such as heat generation and cooling airflow.
3. **Component-Level Special Testing** demonstrates that the equipment meets special constraints it may endure once installed into its designated location. While the "Installed Operation" test ensures that the equipment operates after installation, these tests ensure that the equipment can survive within the confines of the installation space. For example, while general air flow may be assessed during the "Installed Operation" test, it may be known that the equipment may reach extreme temperatures or experience shock or vibrations when installed into the specified area. These special tests validate that the equipment meets the specified operational and non-operational constraints expected within the installation space.

4.2.3.2 Test Level

The test level for StageThree is **Stage-3** as defined by **ref˙˙DOD˙STD˙2106˙NAVY**, **ref˙˙DOD˙STD˙2106˙NAVY** [**ref˙˙DOD˙STD˙2106˙NAVY**].

4.2.3.3 Test Type or Class

The test type or class for StageThree is development testing for installation fit or operation validation and environmental for special testing.

4.2.3.4 Qualification Method

The test qualification method for StageThree generally is performed via test, but analysis may be performed as necessary to ensure each equipment item meets the design specifications.

4.2.3.5 Traceability

The test traceability for StageThree generally is included in the overall **RTVM**. For "Form, Fit, or Function" testing, the results are expected to map to related operational specifications. For "Special Tests", the results are expected to map to the environmental specifications. Both trace classes are expected to map System/Subsystem Specification (**SSS**) implementation requirements to specific tests and/or test cases within a System Test Specification (**STS**).

4.2.3.6 Special Requirements

The test special requirements for StageThree are defined as necessary in test procedures. "Form, Fit, or Function" testing may include references to overall assembly or operational set-up documentation. "Special Testing" details are expected to be performed at a specialized test facility that prepares its own test procedures based upon the facility's own quality management system and certifications.

4.2.3.7 Data Recoding

The test data recoding for StageThree is to record all inspection results in hard copy or electronic form. Electronic form (e.g., a database application with bar code scanning capability) is preferred, but not required, to minimize labor effort and data entry errors. For tests performed by a specialized test facility, they will follow their own processes for data recording and provide records to the development team as appropriate.

4.2.3.8 Assumptions or Constraints

The test assumptions or constraints for StageThree are defined in the associated inspection or test procedures. As noted above, "Special Testing" details are expected to be performed at a specialized test facility.

4.2.3.9 Safety, Security, and Privacy

The test safety, security, and privacy for StageThree are defined in the associated inspection or test procedures. Special care must be taken when test results indicate issues with security measured for the system.

4.2.3.10 Tests and Test Cases

Tests and test cases for this test stage are:

1. Special tests will be determined during the design phase.

4.2.4 Intra-system Tests

This section defines the plans for the StageFour validation efforts.

4.2.4.1 Test Objective

The test objective for StageFour is to demonstrate operations within a system or sub-system. For example, a networking sub-system may be validated to ensure configurations are correct and expected throughput capacities are met. Specialized software or test harnesses may be developed and/or used to perform these tests. This level of testing ensures that the sub-system meets its external interface definitions and functions as expected before being integrated into a larger system of systems.

4.2.4.2 Test Level

The test level for StageFour is **Stage-4** as defined by **ref˙˙DOD˙STD˙2106˙NAVY**, **ref˙˙DOD˙STD˙2106˙NAVY** [**ref˙˙DOD˙STD˙2106˙NAVY**].

4.2.4.3 Test Type or Class

The test type or class for StageFour is development testing. Note that in this case, this testing may also include validation of sub-system external interfaces.

4.2.4.4 Qualification Method

The test qualification method for StageFour generally is performed via test, but analysis may be performed as necessary to ensure each subsystem meets the design specifications.

4.2.4.5 Traceability

The test traceability for StageFour generally is included in the overall **RTVM**. For "Form, Fit, or Function" testing, the results are expected to map to related operational or interface specifications. These trace classes are expected to map System/Subsystem Specification (**SSS**) implementation requirements to specific tests and/or test cases within a System Test Specification (**STS**).

4.2.4.6 Special Requirements

The test special requirements for StageFour are defined as necessary in test procedures. Test procedures within an **STS** may include references to overall assembly or operational set-up documentation to prevent duplication of information.

4.2.4.7 Data Recoding

The test data recoding for StageFour is to record all inspection results in hard copy or electronic form. Electronic form (e.g., a database application with bar code scanning capability) is preferred, but not required, to minimize labor effort and data entry errors.

4.2.4.8 Assumptions or Constraints

The test assumptions or constraints for StageFour are defined in the associated inspection or test procedures.

4.2.4.9 Safety, Security, and Privacy

The test safety, security, and privacy for StageFour are defined in the associated inspection or test procedures. Special care must be taken when test results indicate issues with security measured for the system.

4.2.4.10 Tests and Test Cases

Tests and test cases for this test stage are:

1. **Power** Functional assessment of power distribution and **UPS** capabilities.
   (a) **Power Distribution** Validation of power distribution in Balance System.
   (b) **UPS** Validation of **UPS** in Balance System.
2. **Networking** Functional assessment of network topology and configuration.
   (a) **IP Addressing** Validation of **IPv4** and **IPv6** addressing and routing.
   (b) **NTP Services** Validation of **NTP** services.
3. **Authentication** Functional assessment of user authentication services.
   (a) **Setup** Validation of user and group setup from SysAdmin workstations.
   (b) **Operations** Validation of operational access from all workstations.
4. **Radios** Functional assessment of radio configuration and operation.
   (a) **Configuration** Validation of configuration control from Maintainer workstations.
   (b) **Communication** Validation of operational control from Operator workstations.
   (c) **Arbitration** Validation of arbitration control (operator assignment) from Operator workstations.
5. **Recording** Functional assessment of recording, playback, and offloading capabilities.
   (a) **Recording** Validation of voice recording from Operator workstations.
   (b) **Playback** Validation of voice playback from Maintainer workstation (**KVM**

connected to recorder).

(c) **Offloading** Validation of voice offloading from Maintainer workstation (**KVM** connected to recorder).

4.2.5 Total System Tests

This section defines the plans for the StageFive validation efforts.

4.2.5.1 Test Objective

The test objective for StageFive is to demonstrate operations of the entire Balance System, but with some possible limitations. For example, the entire system may be set up within a laboratory or testbed environment to facilitate **RF** communications between nodes and test endpoints instead of final endpoint targets. The StageFive may also include noise generators so that communications can be stress-tested to demonstrate it meets specified quality of service levels. The goal of this testing to to demonstrate that the system functions as expected in as near-to-deployed scenario as is possible, without being a fully deployed configuration.

4.2.5.2 Test Level

The test level for StageFive is **Stage-5** as defined by **ref˙˙DOD˙STD˙2106˙NAVY**, **ref˙˙DOD˙STD˙2106˙NAVY** [**ref˙˙DOD˙STD˙2106˙NAVY**].

4.2.5.3 Test Type or Class

The test type or class for StageFive is development testing. Note that in this case, this testing may also include validation of system-level external interfaces.

4.2.5.4 Qualification Method

The test qualification method for StageFive is demonstration, inspection, test, or analysis as necessary to ensure each subsystem meets the design specifications.

4.2.5.5 Traceability

The test traceability for StageFive must be included in the overall **RTVM**. For "Form, Fit, or Function" testing, the results are expected to map to related operational or interface specifications. These trace classes are expected to map System/Subsystem Specification (**SSS**) implementation requirements to specific tests and/or test cases within a System Test Specification (**STS**).

4.2.5.6 Special Requirements

The test special requirements for StageFive are defined as necessary in test procedures. Test procedures within an **STS** may include references to overall assembly or operational set-up documentation to prevent duplication of information.

4.2.5.7 Data Recoding

The test data recoding for StageFive is to record all inspection results in hard copy or electronic form. Electronic form (e.g., a database application with bar code scanning capability) is preferred, but not required, to minimize labor effort and data entry errors.

4.2.5.8 Assumptions or Constraints

The test assumptions or constraints for StageFive are defined in the associated inspection or test procedures.

4.2.5.9 Safety, Security, and Privacy

The test safety, security, and privacy for StageFive are defined in the associated inspection or test procedures. Special care must be taken when test results indicate issues with security measured for the system.

4.2.5.10 Tests and Test Cases

Tests and test cases for this test stage are:

1. **Functional assessment of system with ...TBD....**
   (a) **Test Case One** Validation of **...TBD...**.
   (b) **Test Case Two** Validation of **...TBD...**.
2. **...TBD... assessment of system with ...TBD....**
   (a) **Test Case One** Validation of **...TBD...**.
   (b) **Test Case Two** Validation of **...TBD...**.

4.2.6 System Special Tests

This section defines the plans for the StageSix validation efforts.

4.2.6.1 Test Objective

The test objective for StageSix is to demonstrate that the system meets special constraints it may endure in a non-operational state (e.g., during transport) or in an operational state once deployed. This testing may rely on results of Component-Level Special Testing performed during **Stage-3** testing, as defined in § 4.2.3.1 or it may require retesting of the entire system due to transport or operational constraints. Testing between these two levels is required to prevent duplicate assessments.

4.2.6.2 Test Level

The test level for StageSix is **Stage-6** as defined by **ref˙˙DOD˙STD˙2106˙NAVY**, **ref˙˙DOD˙STD˙2106˙NAVY** [**ref˙˙DOD˙STD˙2106˙NAVY**].

4.2.6.3 Test Type or Class

The test type or class for StageSix is environmental testing.

4.2.6.4 Qualification Method

The test qualification method for StageSix generally is performed via test, but analysis may be performed as necessary to ensure the system meets the design specifications.

4.2.6.5 Traceability

The test traceability for StageSix must be included in the overall **RTVM**.

4.2.6.6 Special Requirements

The test special requirements for StageSix are defined as necessary in test procedures. Test procedures within an **STS** may include references to overall assembly or operational set-up documentation to prevent duplication of information. This testing is expected to be performed at a specialized test facility that prepares its own test procedures based upon the facility's own quality management system and certifications.

4.2.6.7 Data Recoding

The test data recoding for StageSix is to record all inspection results in hard copy or electronic form. Electronic form (e.g., a database application with bar code scanning capability) is preferred, but not required, to minimize labor effort and data entry errors. For tests performed by a specialized test facility, they will follow their own processes for data recording and provide records to the development team as appropriate.

4.2.6.8 Assumptions or Constraints

The test assumptions or constraints for StageSix are defined in the associated inspection or test procedures. These details are expected to be performed at a specialized test facility.

4.2.6.9 Safety, Security, and Privacy

The test safety, security, and privacy for StageSix are defined in the associated inspection or test procedures. Special care must be taken when test results indicate issues with security measured for the system.

4.2.6.10 Tests and Test Cases

Tests and test cases for this test stage are:

1. **...TBD...** Special tests will be determined during the design phase.

4.2.7 Total Integrated System Tests

The Stage-7 testing is divided into two sections: **DT** and **OT**.

4.2.7.1 Design Testing

This section defines the plans for the **DT** validation efforts.

4.2.7.1.1 Test Objective

The test objective for **DT** is to demonstrate that the entire Balance System meets the approved design specifications, with all tests performed by an outside agency. The goal of this testing is to demonstrate that the system provides all capabilities in a deployed scenario, with all external interfaces being supplied by the actual systems.

4.2.7.1.2 Test Level

The test level for **DT** is **Stage-7** as defined by **ref˙˙DOD˙STD˙2106˙NAVY**, **ref˙˙DOD˙STD˙2106˙**

4.2.7.1.3 Test Type or Class

The test type or class for **DT** is design testing (**DT**).

4.2.7.1.4 Qualification Method

The test qualification method for **DT** generally is performed via test, but analysis may be performed as necessary to ensure the system meets the design specifications.

4.2.7.1.5 Traceability

The test traceability for **DT** must be included in the overall **RTVM**.

4.2.7.1.6 Special Requirements

The test special requirements for **DT** are defined as necessary in test procedures. Test procedures within an **STS** may include references to overall assembly or operational set-up documentation to prevent duplication of information. This testing is expected to be performed by a separate test organization that may prepare its own test procedures based upon the organization's own quality management system and certifications.

4.2.7.1.7 Data Recoding

The test data recoding for **DT** is to record all inspection results in hard copy or electronic form. Electronic form (e.g., a database application with bar code scanning capability) is preferred, but not required, to minimize labor effort and data entry errors. For tests performed by a separate test organization, they will follow their own processes for data recording and provide records to the development team as appropriate.

4.2.7.1.8 Assumptions or Constraints

The test assumptions or constraints for **DT** are defined in the associated inspection or test procedures. These details are expected to be performed by a separate test organization.

4.2.7.1.9 Safety, Security, and Privacy

The test safety, security, and privacy for **DT** are defined in the associated inspection or test procedures. Special care must be taken when test results indicate issues with security measured for the system.

4.2.7.2 Tests and Test Cases

Tests and test cases for this test stage are:

1. **...TBD...** Design qualification tests will be determined by the **DT** organization.

4.2.7.3 Operational Testing

This section defines the plans for the **OT** validation efforts.

4.2.7.3.1 Test Objective

The test objective for **OT** is to demonstrate operations of the entire Balance System, with all tests performed by an outside agency. The goal of this testing is to demonstrate that the system functions as expected in a deployed scenario, with all external interfaces being supplied by the actual systems.

4.2.7.3.2 Test Level

The test level for **OT** is **Stage-7** as defined by **ref˙˙DOD˙STD˙2106˙NAVY**, **ref˙˙DOD˙STD˙2106˙**

4.2.7.3.3 Test Type or Class

The test type or class for **OT** is operational testing (**OT**).

4.2.7.3.4 Qualification Method

The test qualification method for **OT** generally is performed via demonstration, but other methods may be performed as necessary to ensure the system meets the operational specifications.

4.2.7.3.5 Traceability

The test traceability for **OT** must be included in the overall **RTVM**.

4.2.7.3.6 Special Requirements

The test special requirements for **OT** are defined as necessary in test procedures. Test procedures within an **STS** may include references to overall assembly or operational set-up documentation to prevent duplication of information. This testing is expected to be performed by a separate test organization that may prepare its own test procedures based upon the organization's own quality management system and certifications.

4.2.7.3.7 Data Recoding

The test data recoding for **OT** is to record all inspection results in hard copy or electronic form. Electronic form (e.g., a database application with bar code scanning capability) is preferred, but not required, to minimize labor effort and data entry errors. For tests performed by a separate test organization, they will follow their own processes for data recording and provide records to the development team as appropriate.

4.2.7.3.8 Assumptions or Constraints

The test assumptions or constraints for **OT** are defined in the associated inspection or test procedures. These details are expected to be performed by a separate test organization.

4.2.7.3.9 Safety, Security, and Privacy

The test safety, security, and privacy for **OT** are defined in the associated inspection or test procedures. Special care must be taken when test results indicate issues with security measured for the system.

4.2.7.4 Tests and Test Cases

Tests and test cases for this test stage are:

1. **...TBD...** Operational qualification tests will be determined by the **OT** organization.

CHAPTER 5

Schedule

This section provides an overview of the testing schedule.

When the **STP** *is developed, a general time line should be established so general dates should be understood, but exact dates may not be known. Thus, this schedule sets expectations for need-by dates for resources such as laboratories, test ranges, etc.*

CHAPTER 6

Traceability

This section provides traceability of the system components and interfaces to the design requirements.

CHAPTER 7

Notes

This chapter is for testing.

7.1 Note Area 1

This section is for testing.

APPENDIX

Additional Information

This section provides additional information, as necessary, to augment the **STP**.

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